

AI-Agentive Triage and Investigation Engine

Given the shift toward the "**Agentive SOC**," employers value candidates who can supervise and build autonomous security workflows 3, 4. This project involves creating a system that uses **Large Language Models (LLMs)** to independently execute investigative playbooks, such as pulling identity context, querying SIEM history, and checking EDR telemetry before a human analyst even opens a ticket 5. Your tool should correlate event logs across multiple platforms to construct a unified incident timeline and generate a **natural language summary** for human review 5, 6.

IoT Security and Firmware Vulnerability Lab

As the attack surface expands to billions of connected devices, **IoT security** specialists are in high demand 13, 14. Set up a home lab to perform **APK decompilation** and firmware analysis on connected hardware to identify vulnerabilities like unencrypted communication protocols or hardcoded credentials 15, 16. Documenting a full **hardening guide** for a specific hardware target demonstrates the hands-on "builder" mindset that distinguishes security engineers from entry-level analysts 17, 18.

Portfolio Presentation Strategy

Big Tech recruiters look for the **XYZ format** to quantify impact: explain how you "Accomplished **X** through **Y**, resulting in **Z** impact" (e.g., "Automated a manual 4-hour investigation process to take 5 minutes") 30, 31. Ensure every project includes a **polished README** that explains one critical architecture decision and links to a **live endpoint** or a demo video 32, 33.